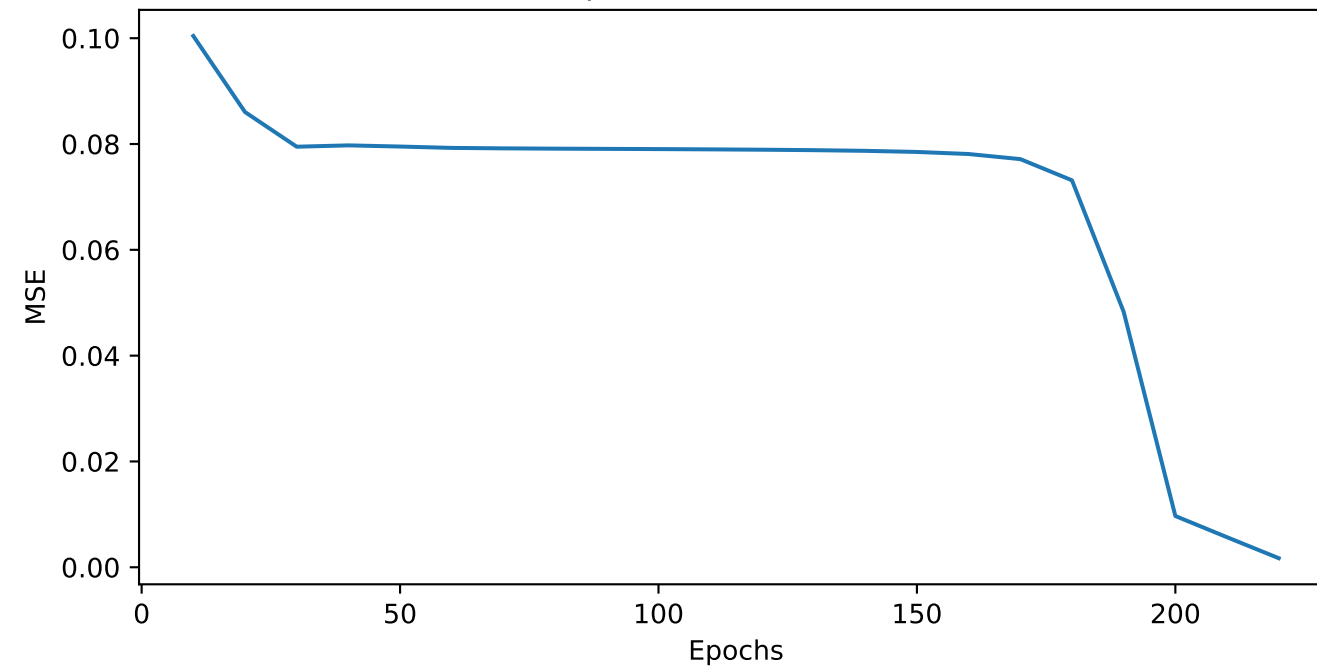
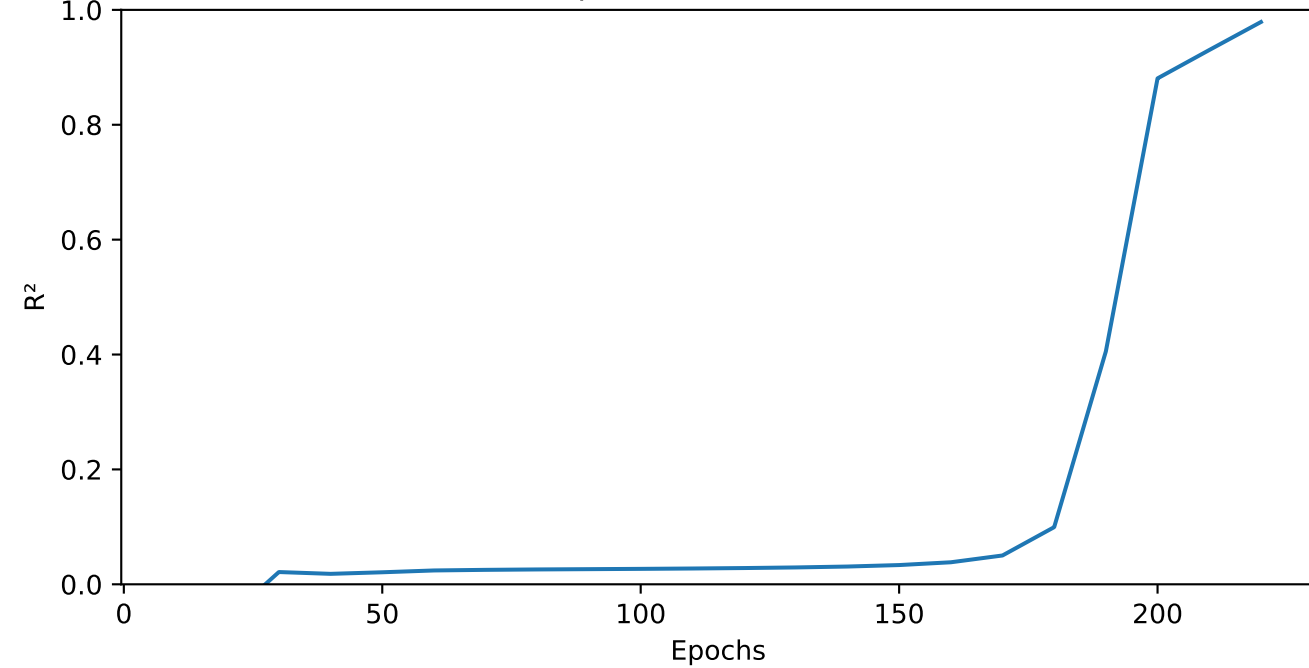


# Neural Network Regression Report - dane1.txt

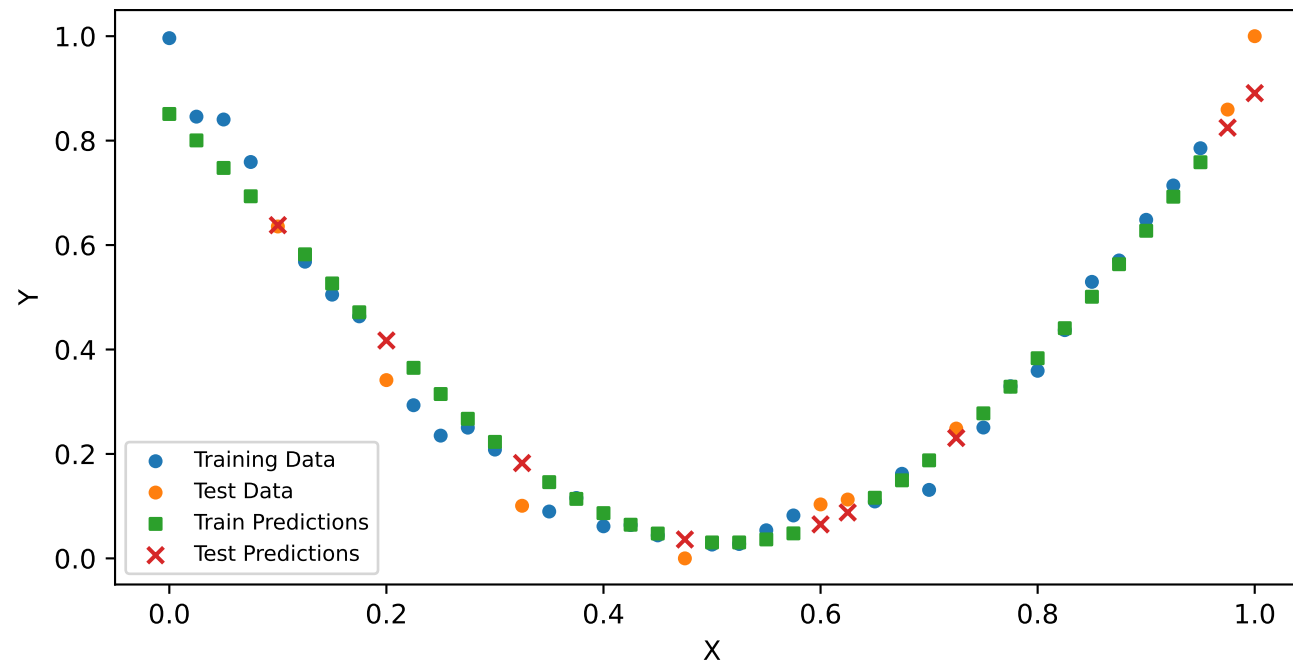
MSE vs Epochs, Final MSE = 0.00192



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9764



Real vs Predicted



Dataset: dane1.txt

Architecture:  
Input neurons: 1  
Hidden neurons: 80 - 80  
Output neurons: 1  
Total parameters: 6721

Training:  
Epochs used: 220  
Learning rate: 0.005  
Regularisation lambda: 1e-07

Activation:  
Function: sine  
Derivative: implemented in code

Initialisation:  
Weights: xavier  
Biases: zeros

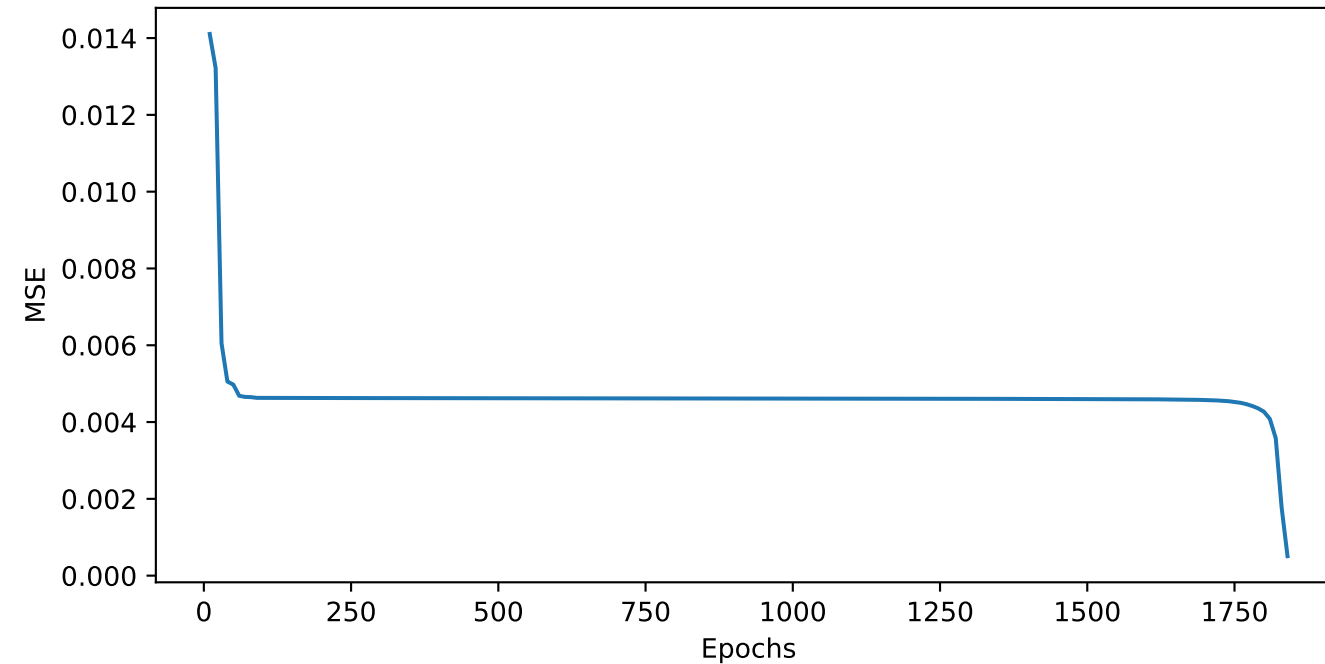
Optimizer:  
Adam optimizer implemented from scratch

Final scores:  
Train MSE: 0.001916  
Train R<sup>2</sup>: 0.976415

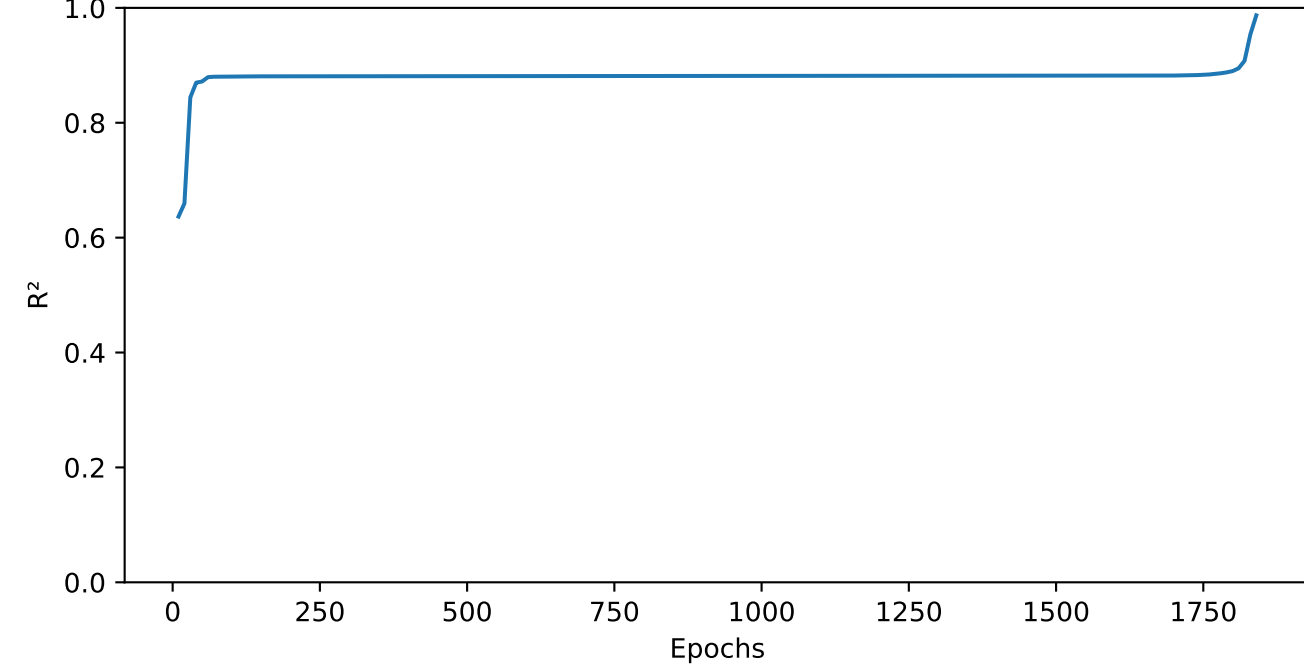
Test MSE: 0.003250  
Test R<sup>2</sup>: 0.972631

# Neural Network Regression Report - dane2.txt

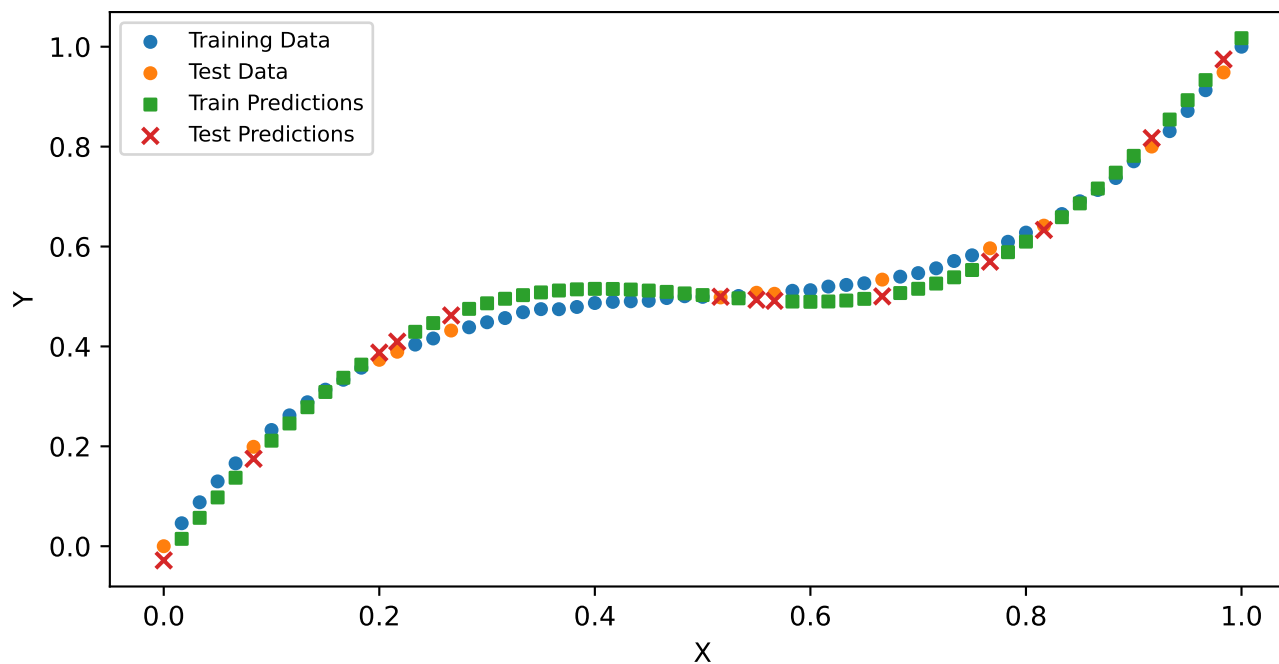
MSE vs Epochs, Final MSE = 0.00062



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9840



Real vs Predicted



Dataset: dane2.txt

Architecture:

Input neurons: 1

Hidden neurons: 80 - 80

Output neurons: 1

Total parameters: 6721

Training:

Epochs used: 1840

Learning rate: 0.005

Regularisation lambda: 1e-07

Activation:

Function: sine

Derivative: implemented in code

Initialisation:

Weights: xavier

Biases: zeros

Optimizer:

Adam optimizer implemented from scratch

Final scores:

Train MSE: 0.000620

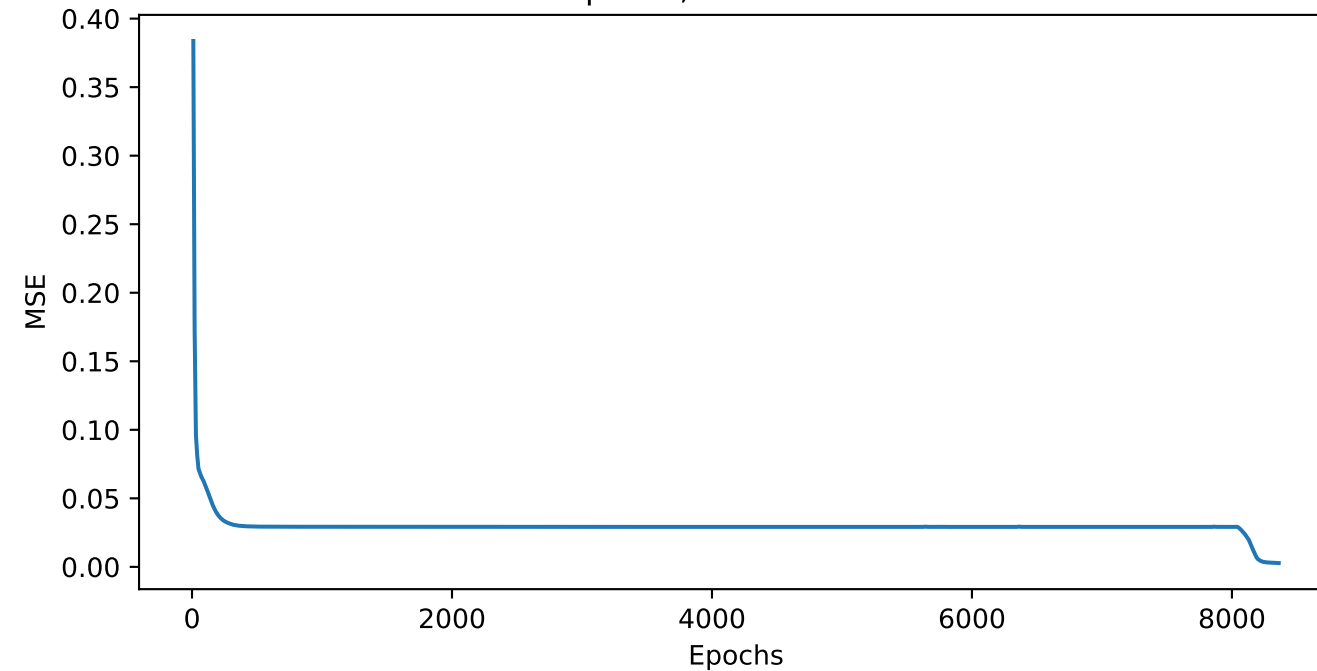
Train R<sup>2</sup>: 0.984046

Test MSE: 0.000476

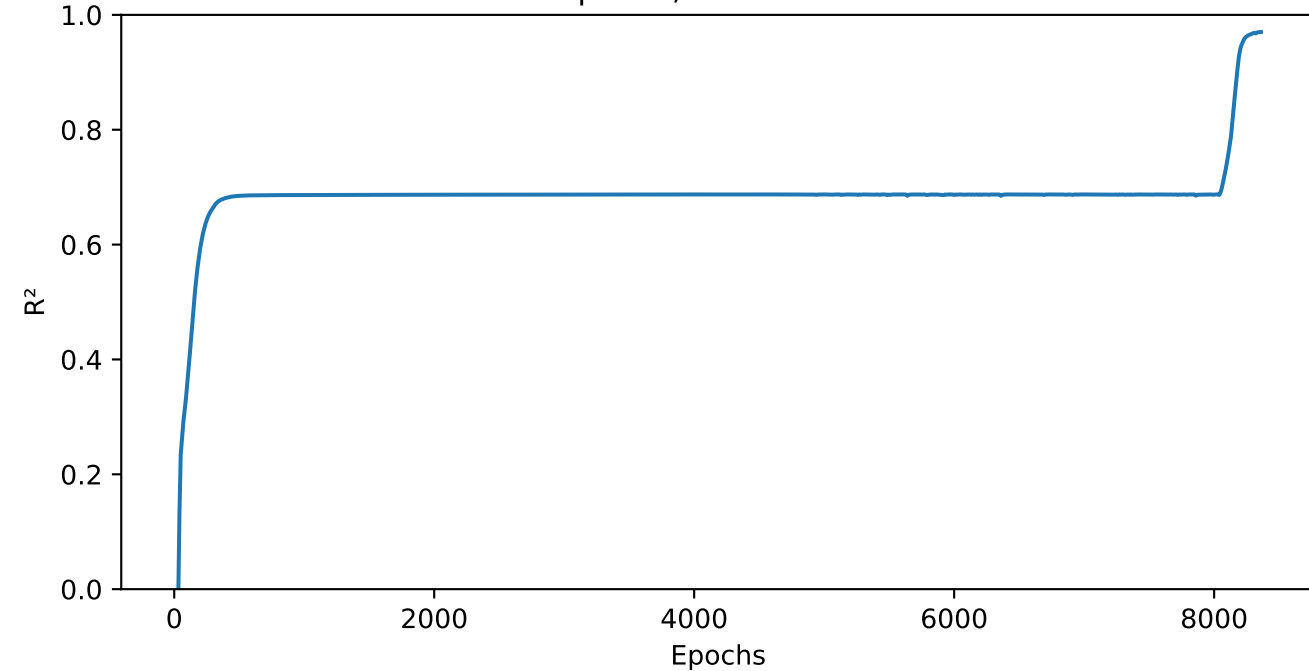
Test R<sup>2</sup>: 0.991102

# Neural Network Regression Report - dane3.txt

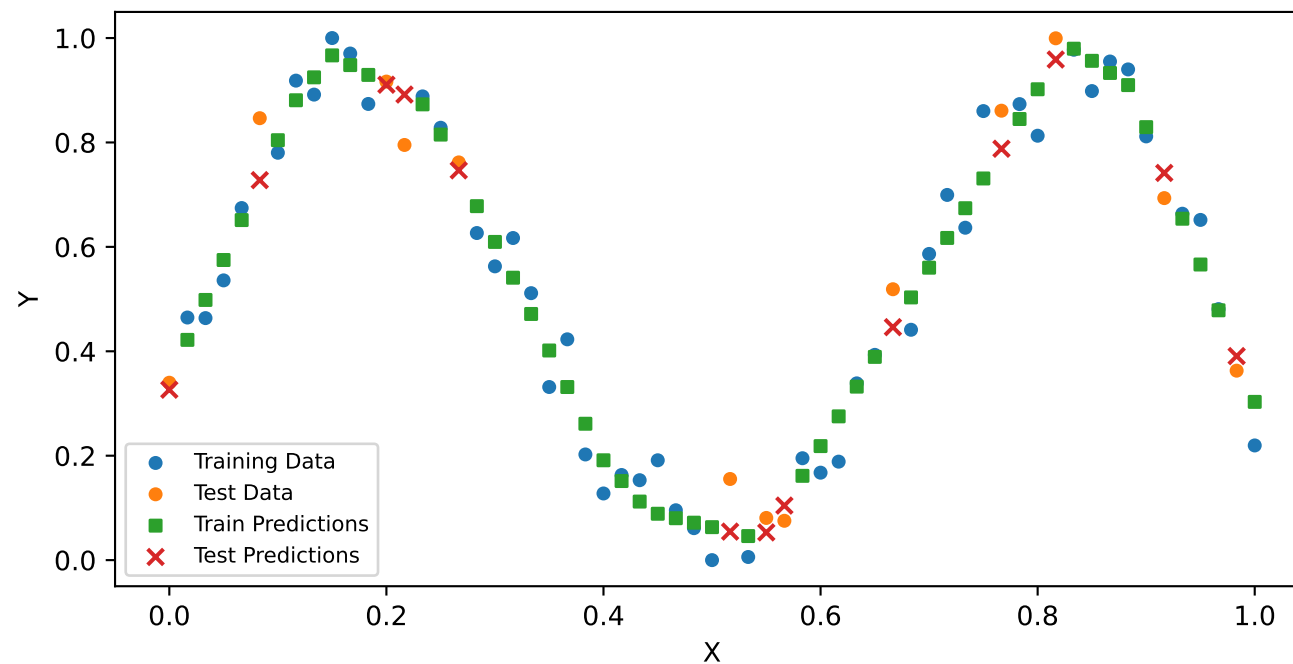
MSE vs Epochs, Final MSE = 0.00278



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9702



Real vs Predicted



Dataset: dane3.txt

Architecture:  
Input neurons: 1  
Hidden neurons: 120  
Output neurons: 1  
Total parameters: 361

Training:  
Epochs used: 8360  
Learning rate: 0.005  
Regularisation lambda: 1e-05

Activation:  
Function: relu  
Derivative: implemented in code

Initialisation:  
Weights: he  
Biases: small\_positive

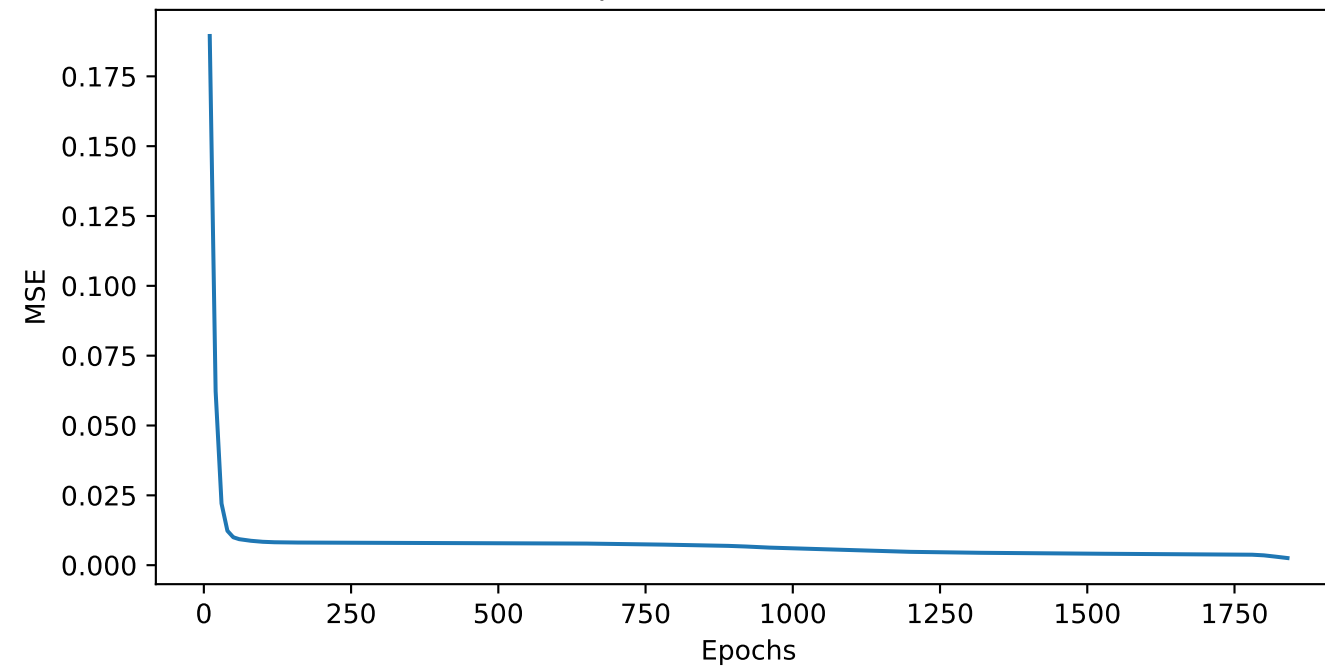
Optimizer:  
Adam optimizer implemented from scratch

Final scores:  
Train MSE: 0.002777  
Train R<sup>2</sup>: 0.970222

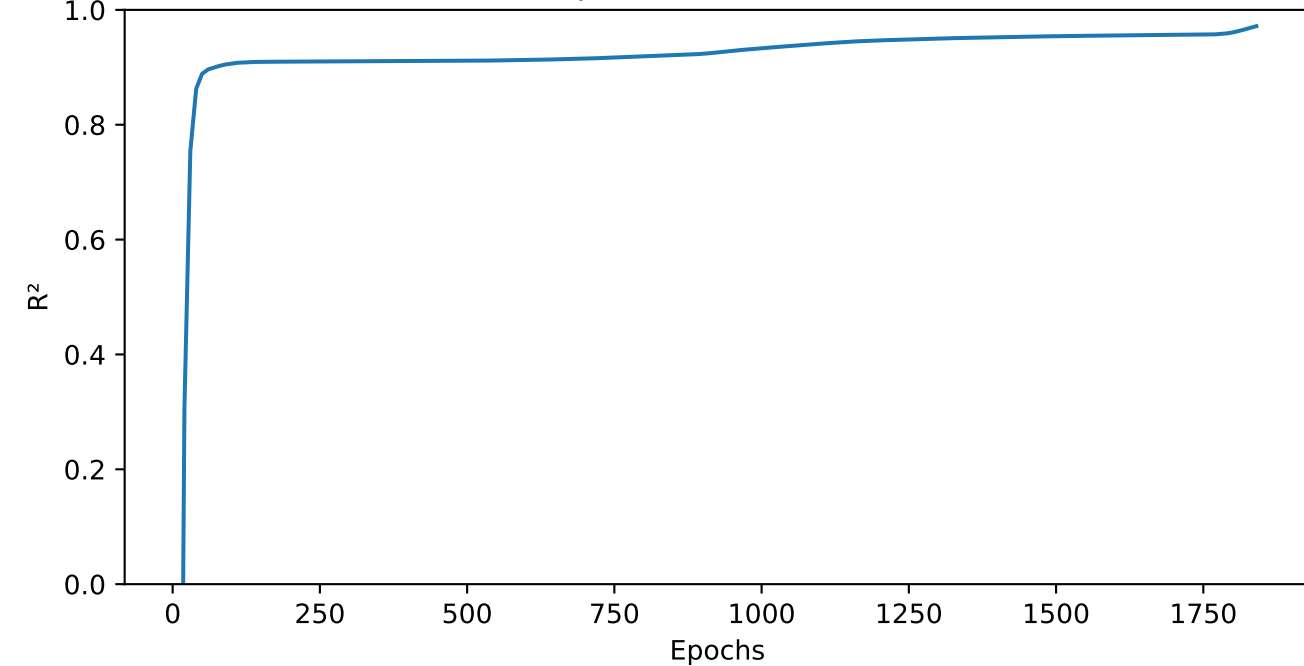
Test MSE: 0.003921  
Test R<sup>2</sup>: 0.961467

# Neural Network Regression Report - dane4.txt

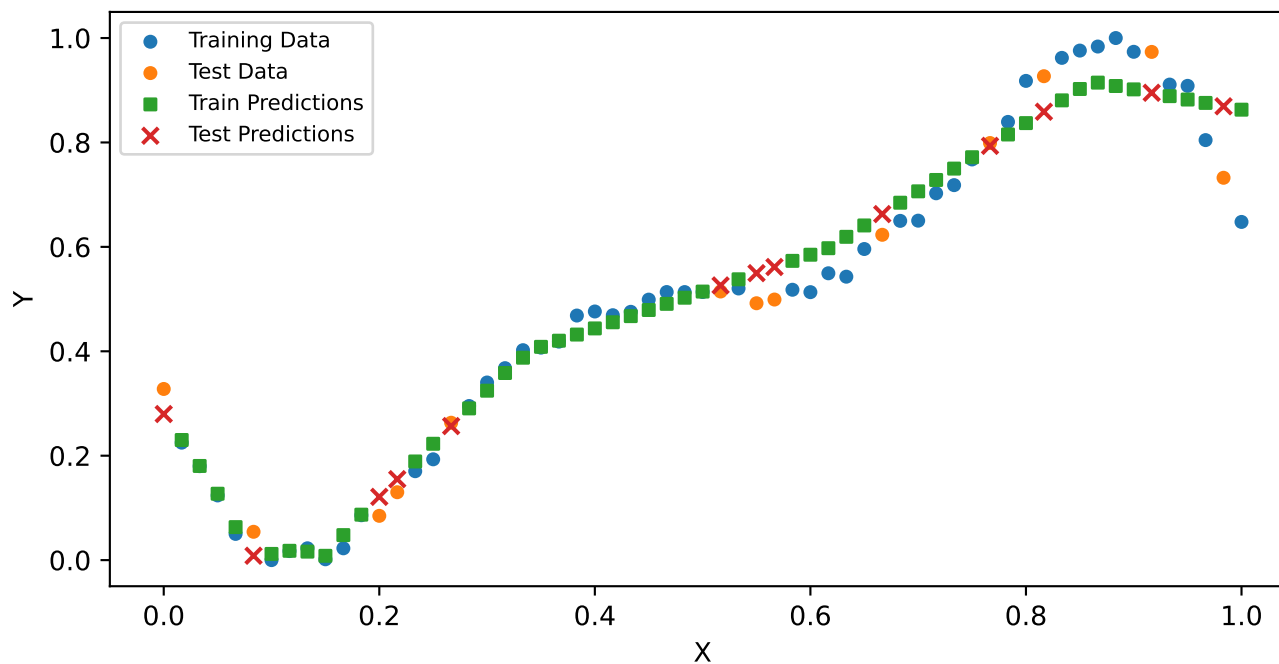
MSE vs Epochs, Final MSE = 0.00253



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9718



Real vs Predicted



Dataset: dane4.txt

Architecture:

Input neurons: 1

Hidden neurons: 120

Output neurons: 1

Total parameters: 361

Training:

Epochs used: 1840

Learning rate: 0.005

Regularisation lambda: 1e-05

Activation:

Function: relu

Derivative: implemented in code

Initialisation:

Weights: he

Biases: small\_positive

Optimizer:

Adam optimizer implemented from scratch

Final scores:

Train MSE: 0.002527

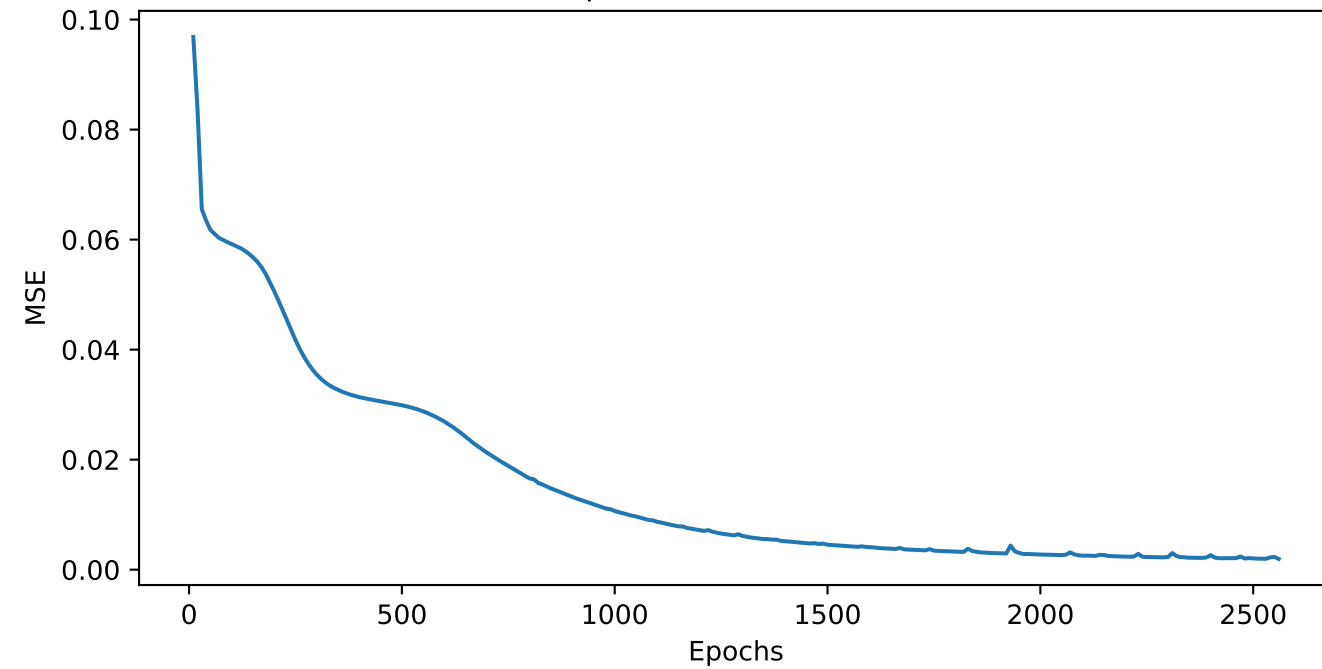
Train R<sup>2</sup>: 0.971823

Test MSE: 0.003450

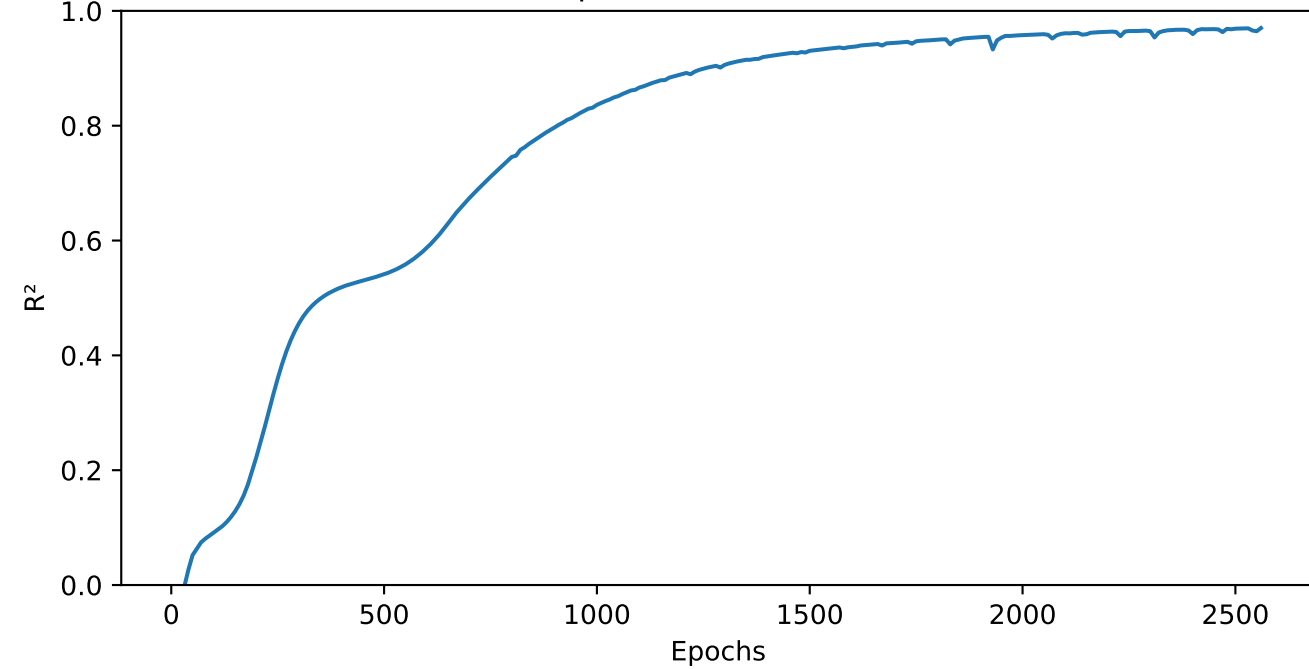
Test R<sup>2</sup>: 0.961282

# Neural Network Regression Report - dane5.txt

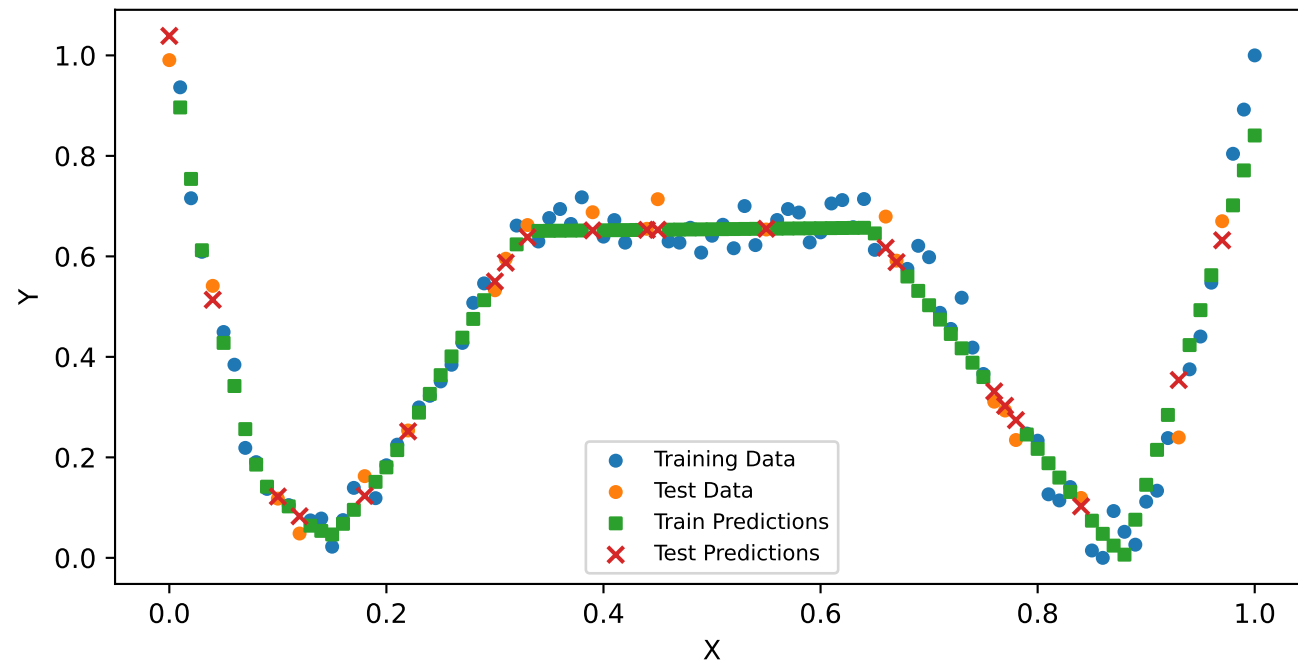
MSE vs Epochs, Final MSE = 0.00202



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9690



Real vs Predicted



Dataset: dane5.txt

Architecture:  
Input neurons: 1  
Hidden neurons: 100  
Output neurons: 1  
Total parameters: 301

Training:  
Epochs used: 2560  
Learning rate: 0.005  
Regularisation lambda: 1e-05

Activation:  
Function: leaky\_relu  
Derivative: implemented in code

Initialisation:  
Weights: he  
Biases: small\_positive

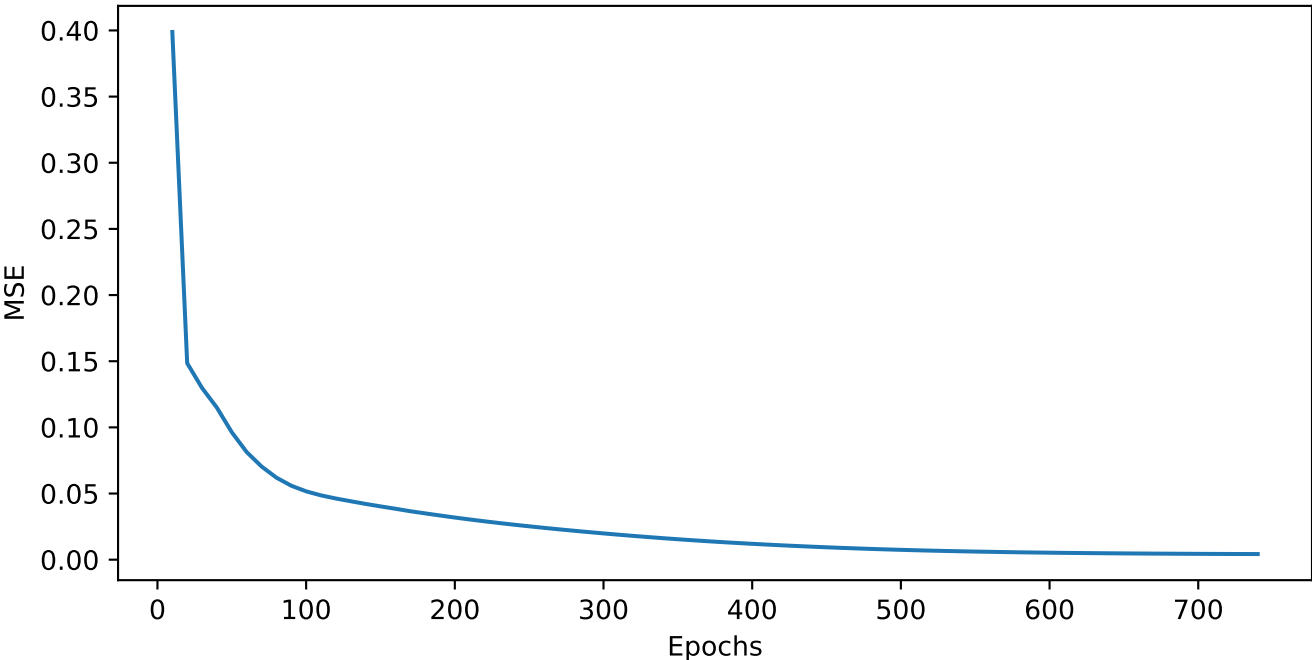
Optimizer:  
Adam optimizer implemented from scratch

Final scores:  
Train MSE: 0.002021  
Train R<sup>2</sup>: 0.968980

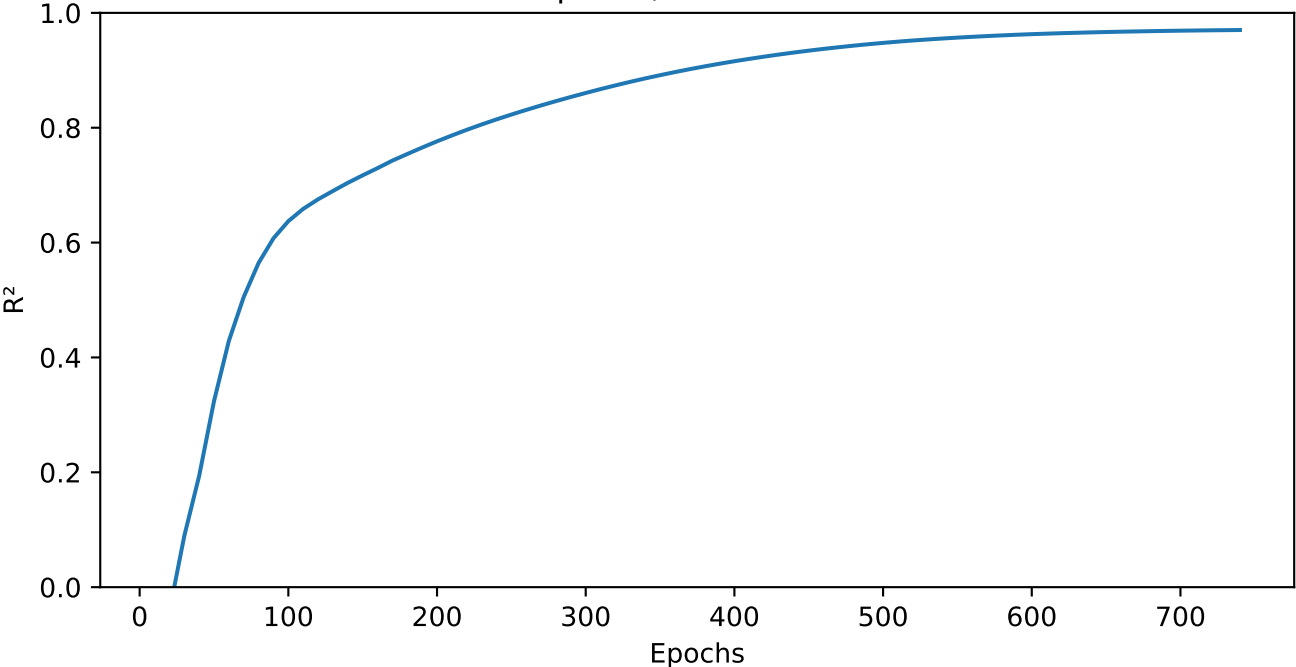
Test MSE: 0.001549  
Test R<sup>2</sup>: 0.975656

# Neural Network Regression Report - dane6.txt

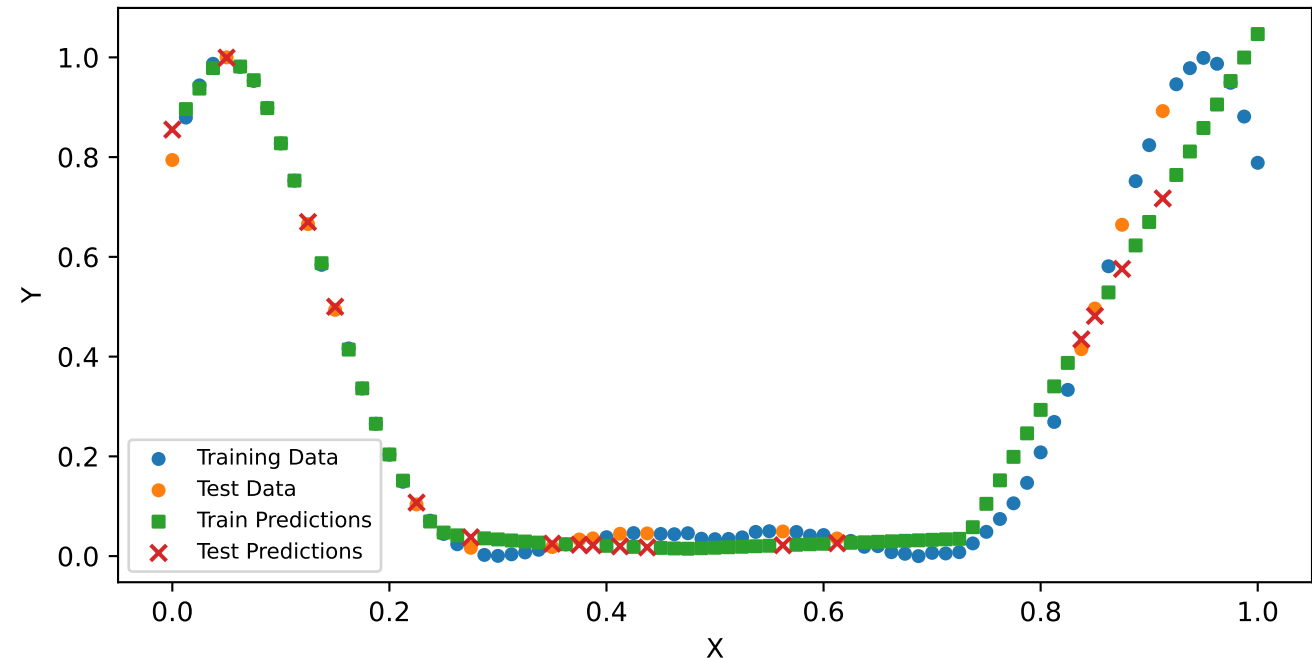
MSE vs Epochs, Final MSE = 0.00425



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9702



Real vs Predicted



Dataset: dane6.txt

Architecture:

Input neurons: 1

Hidden neurons: 120

Output neurons: 1

Total parameters: 361

Training:

Epochs used: 740

Learning rate: 0.005

Regularisation lambda: 1e-05

Activation:

Function: relu

Derivative: implemented in code

Initialisation:

Weights: he

Biases: small\_positive

Optimizer:

Adam optimizer implemented from scratch

Final scores:

Train MSE: 0.004249

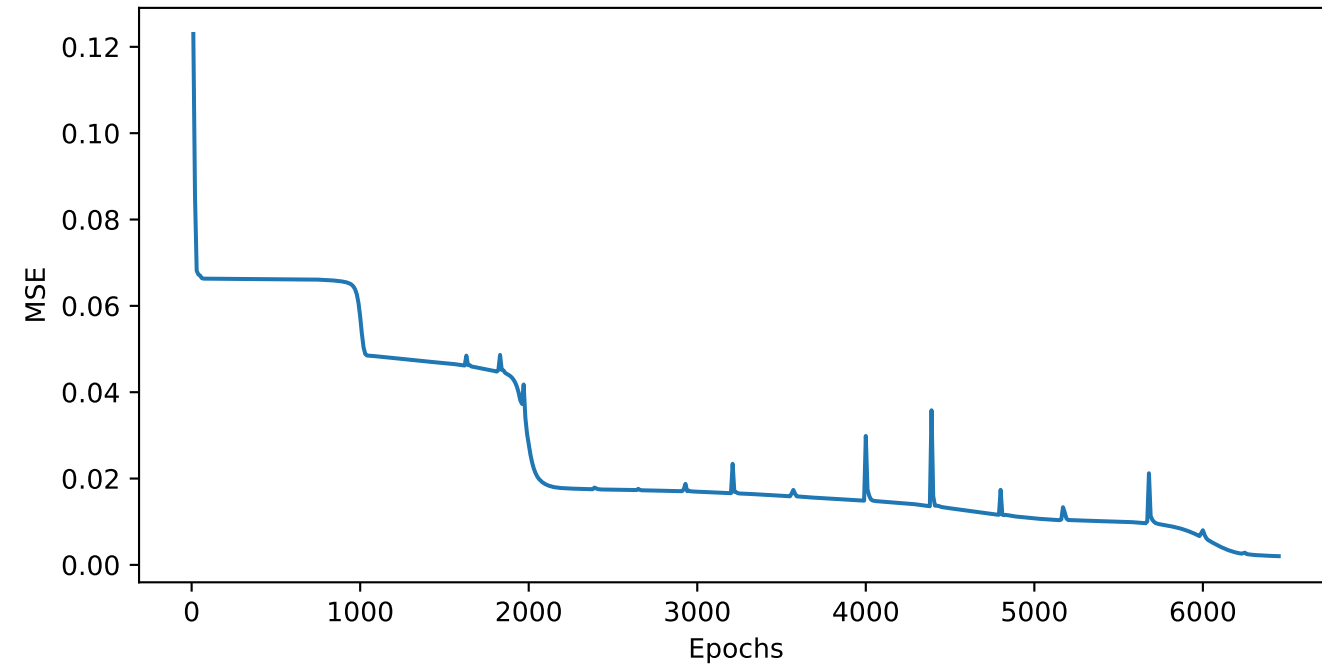
Train R<sup>2</sup>: 0.970171

Test MSE: 0.002706

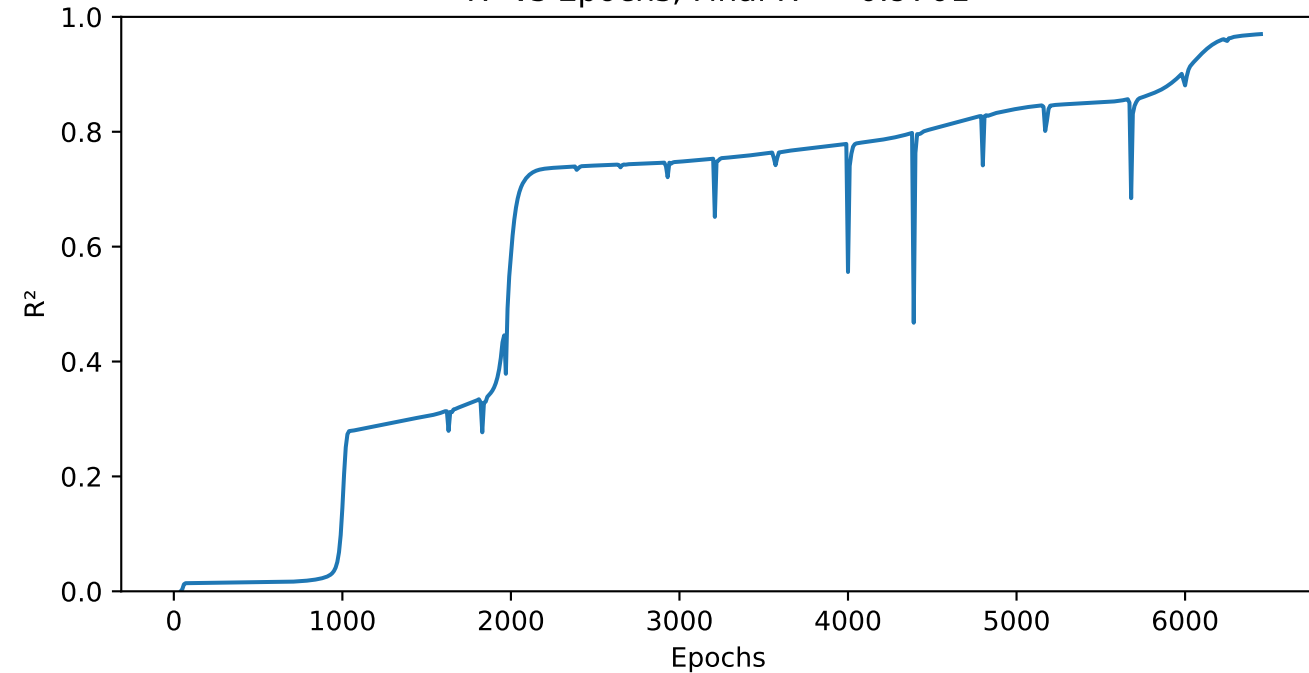
Test R<sup>2</sup>: 0.977155

# Neural Network Regression Report - dane7.txt

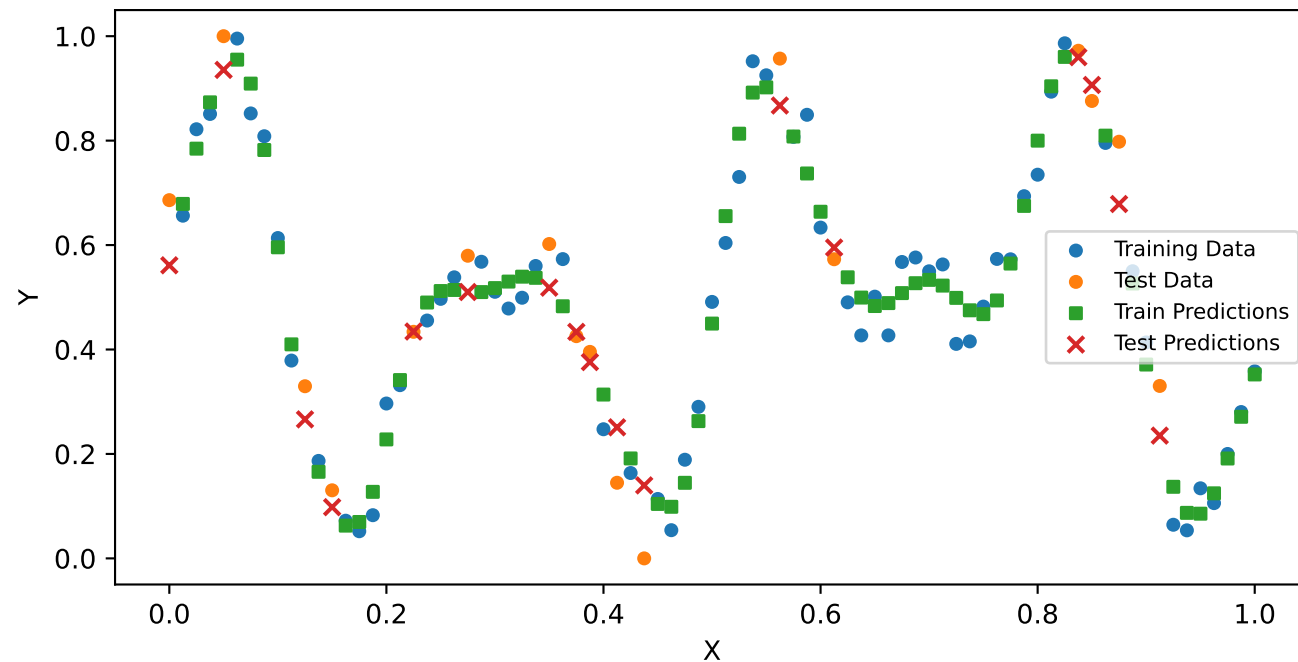
MSE vs Epochs, Final MSE = 0.00201



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9701



Real vs Predicted



Dataset: dane7.txt

Architecture:  
Input neurons: 1  
Hidden neurons: 120 - 120  
Output neurons: 1  
Total parameters: 14881

Training:  
Epochs used: 6450  
Learning rate: 0.003  
Regularisation lambda: 1e-06

Activation:  
Function: tanh  
Derivative: implemented in code

Initialisation:  
Weights: xavier  
Biases: zeros

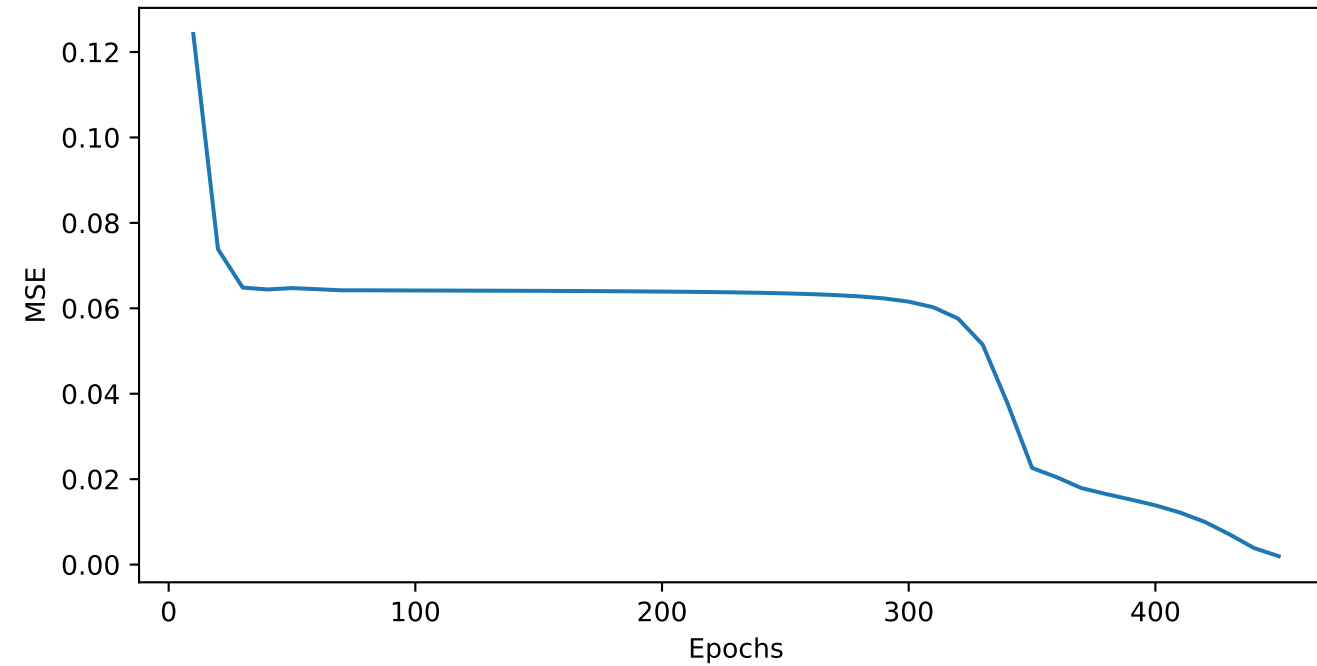
Optimizer:  
Adam optimizer implemented from scratch

Final scores:  
Train MSE: 0.002014  
Train R<sup>2</sup>: 0.970053

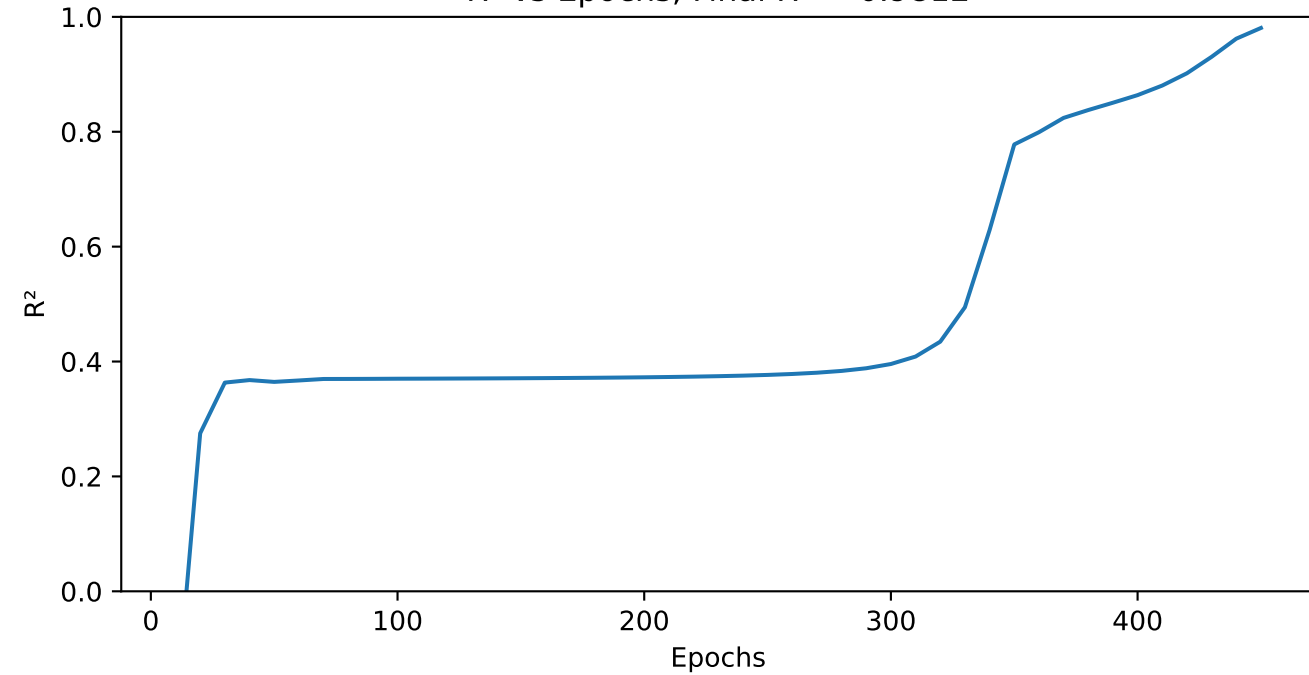
Test MSE: 0.005931  
Test R<sup>2</sup>: 0.934031

# Neural Network Regression Report - dane8.txt

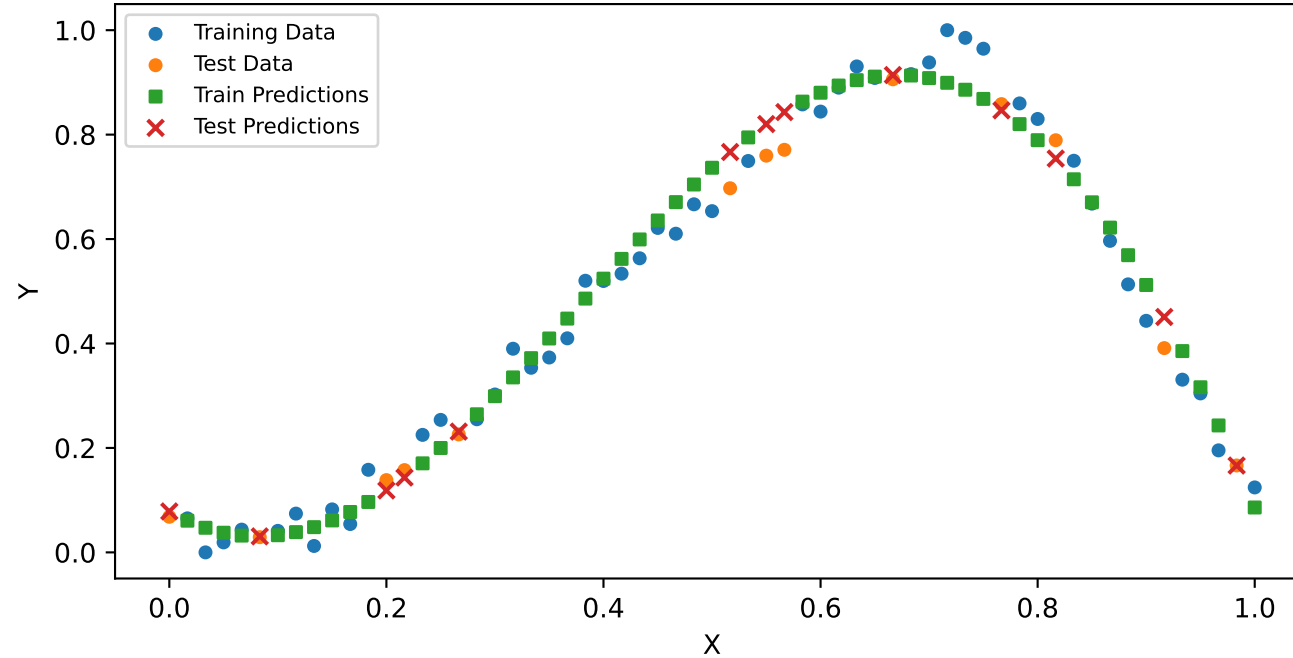
MSE vs Epochs, Final MSE = 0.00191



R<sup>2</sup> vs Epochs, Final R<sup>2</sup> = 0.9812



Real vs Predicted



Dataset: dane8.txt

Architecture:

Input neurons: 1

Hidden neurons: 150 - 150

Output neurons: 1

Total parameters: 23101

Training:

Epochs used: 450

Learning rate: 0.003

Regularisation lambda: 1e-07

Activation:

Function: sine

Derivative: implemented in code

Initialisation:

Weights: xavier

Biases: zeros

Optimizer:

Adam optimizer implemented from scratch

Final scores:

Train MSE: 0.001912

Train R<sup>2</sup>: 0.981229

Test MSE: 0.001494

Test R<sup>2</sup>: 0.986017

Summary of all datasets

| Dataset   | Activation | Layers           | Init   | LR    | Lambda | Epochs | Params | Test MSE | Test R²  |
|-----------|------------|------------------|--------|-------|--------|--------|--------|----------|----------|
| dane1.txt | sine       | [1, 80, 80, 1]   | xavier | 0.005 | 1e-07  | 220    | 6721   | 0.003250 | 0.972631 |
| dane2.txt | sine       | [1, 80, 80, 1]   | xavier | 0.005 | 1e-07  | 1840   | 6721   | 0.000476 | 0.991102 |
| dane3.txt | relu       | [1, 120, 1]      | he     | 0.005 | 1e-05  | 8360   | 361    | 0.003921 | 0.961467 |
| dane4.txt | relu       | [1, 120, 1]      | he     | 0.005 | 1e-05  | 1840   | 361    | 0.003450 | 0.961282 |
| dane5.txt | leaky_relu | [1, 100, 1]      | he     | 0.005 | 1e-05  | 2560   | 301    | 0.001549 | 0.975656 |
| dane6.txt | relu       | [1, 120, 1]      | he     | 0.005 | 1e-05  | 740    | 361    | 0.002706 | 0.977155 |
| dane7.txt | tanh       | [1, 120, 120, 1] | xavier | 0.003 | 1e-06  | 6450   | 14881  | 0.005931 | 0.934031 |
| dane8.txt | sine       | [1, 150, 150, 1] | xavier | 0.003 | 1e-07  | 450    | 23101  | 0.001494 | 0.986017 |